

EXPERIMENT 1

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A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE ,BIVARIATE AND MULTIVARIATE REGRESSION

AIM:

TO IMPLEMENT A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE
,BIVARIATE AND MULTIVARIATE REGRESSION

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
df = pd.read_csv('./input/iris-dataset/iris.csv') df.head(150)
df.shape (150,5)

#univariate for sepal width
df.loc[df['variety']=='Setosa'] df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['sepal.width'],np.zeros_like(df_Setosa['sepal.widt
h']))
plt.scatter(df_Virginica['sepal.width'],np.zeros_like(df_Virginica['sepal.
width']))
plt.scatter(df_Versicolor['sepal.width'],np.zeros_like(df_Versicolor['sep
al.width']))
plt.xlabel('sepal.width')
plt.show()

#univariate for sepal length
df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
```

```
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['sepal.length'],np.zeros_like(df_Setosa['sepal.length']))
plt.scatter(df_Virginica['sepal.length'],np.zeros_like(df_Virginica['sepal.length']))
plt.scatter(df_Versicolor['sepal.length'],np.zeros_like(df_Versicolor['sepal.length'])) plt.xlabel('sepal.length')
plt.show()
```

#univariate for petal width

```
df.loc[df['variety']=='Setosa']
df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['petal.width'],np.zeros_like(df_Setosa['petal.width']))
plt.scatter(df_Virginica['petal.width'],np.zeros_like(df_Virginica['petal.width']))
plt.scatter(df_Versicolor['petal.width'],np.zeros_like(df_Versicolor['petal.width']))
plt.xlabel('petal.width')
plt.show()
```

#univariate for petal length

```
df.loc[df['variety']=='Setosa'] df_Setosa=df.loc[df['variety']=='Setosa']
df_Virginica=df.loc[df['variety']=='Virginica']
df_Versicolor=df.loc[df['variety']=='Versicolor']
plt.scatter(df_Setosa['petal.length'],np.zeros_like(df_Setosa['petal.length']))
plt.scatter(df_Virginica['petal.length'],np.zeros_like(df_Virginica['petal.length']))
plt.scatter(df_Versicolor['petal.length'],np.zeros_like(df_Versicolor['petal.length'])) plt.xlabel('petal.length')
plt.show()
```

#bivariate sepal.width vs petal.width

```
sns.FacetGrid(df,hue='variety',height=5).map(plt.scatter,"sepal.width","  
petal.width").add_legend();  
plt.show()
```

#bivariate sepal.length vs petal.length

```
sns.FacetGrid(df,hue='variety',height=5).map(plt.scatter,"sepal.length",  
"petal.length").add_legend();  
plt.show()
```

#multivariate all the features

```
sns.pairplot(df,hue="variety",size=2)
```

OUTPUT:

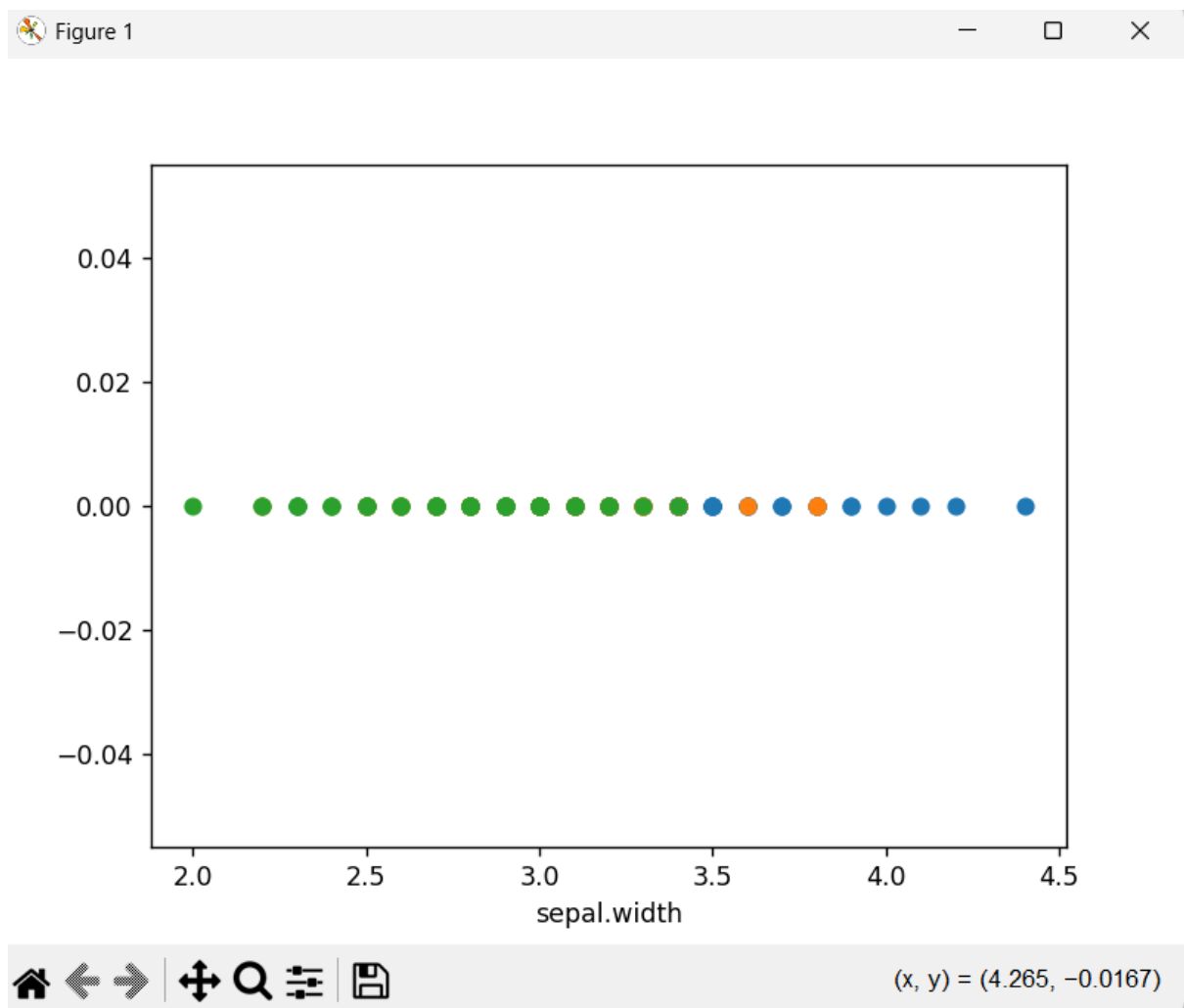
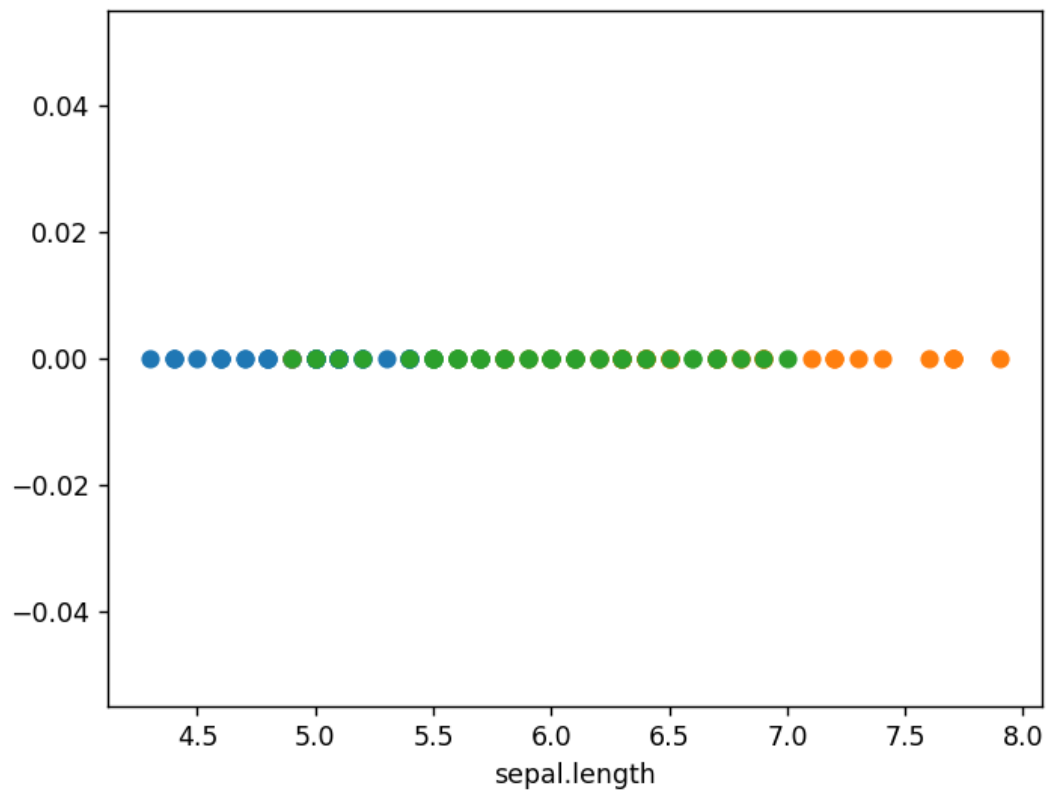




Figure 1



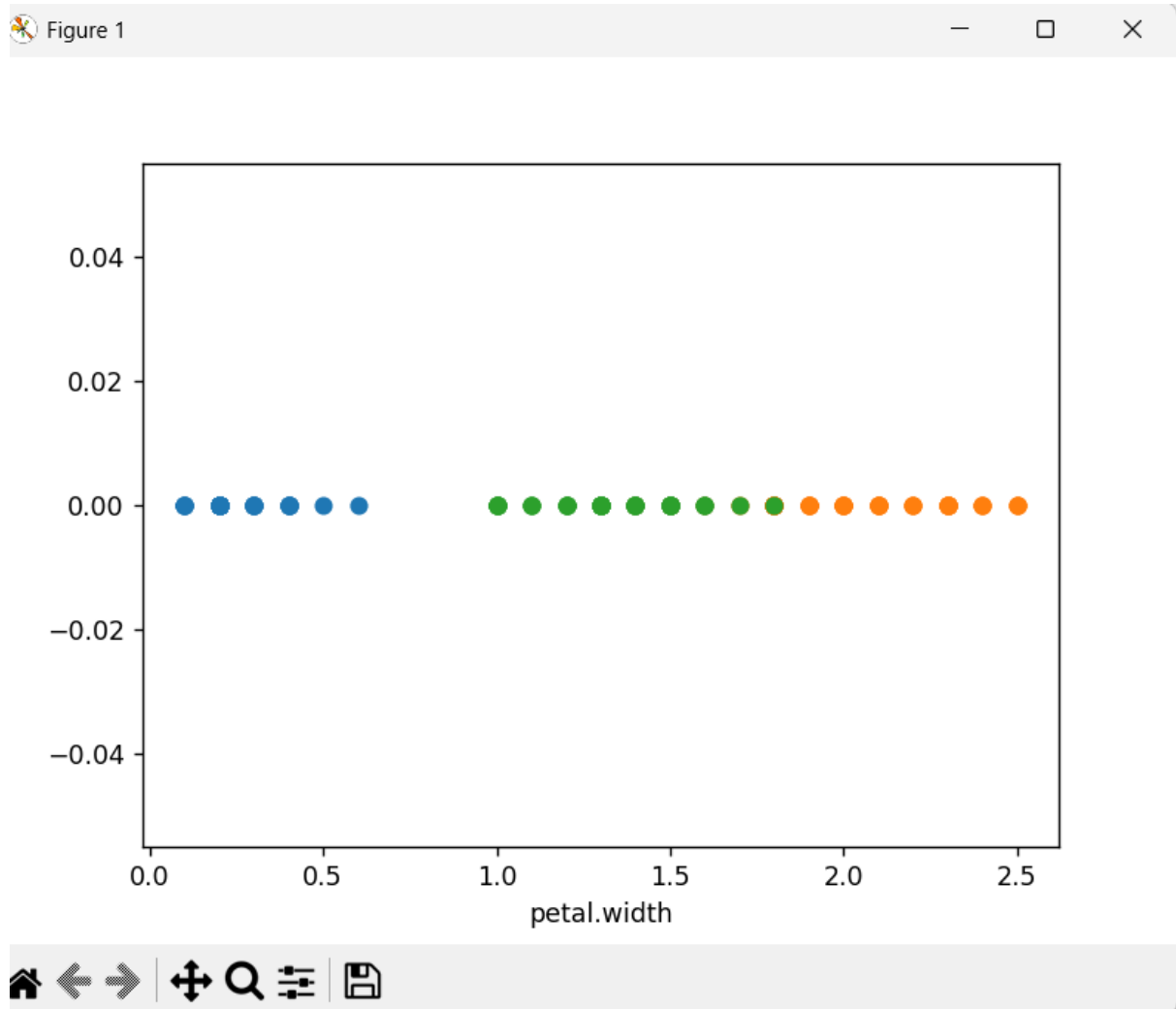
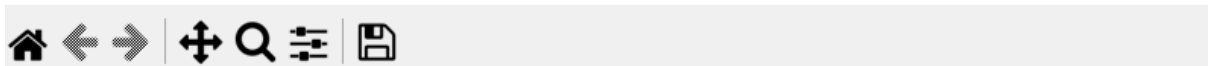
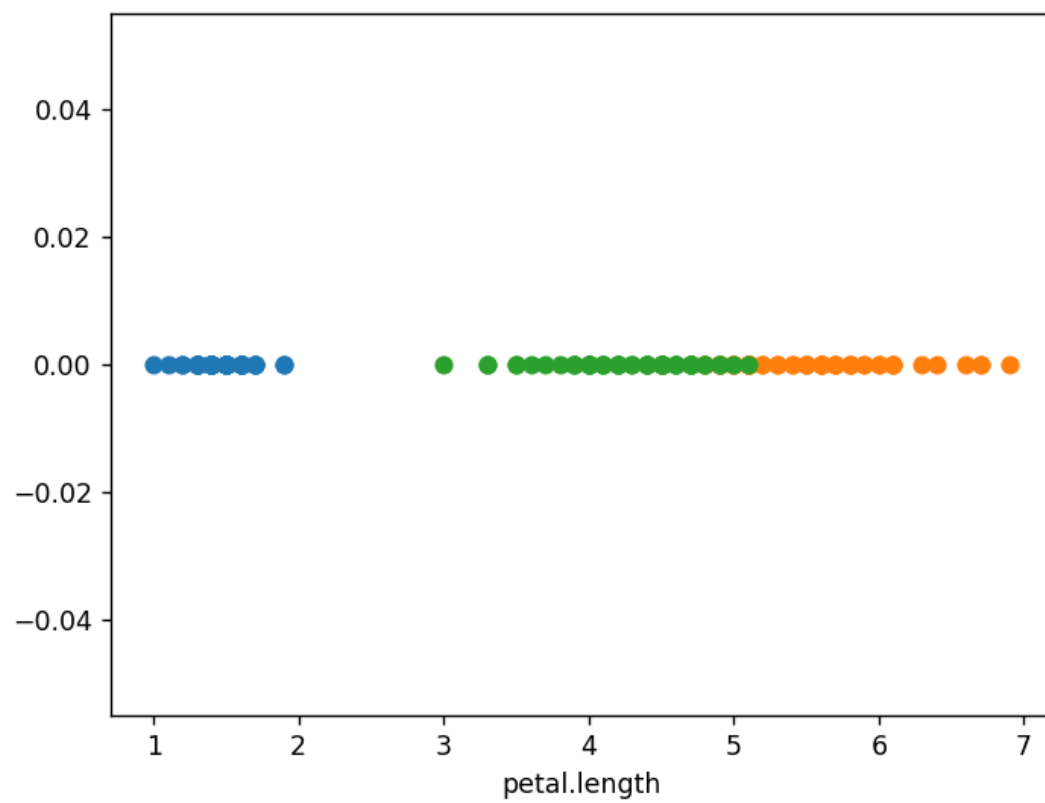
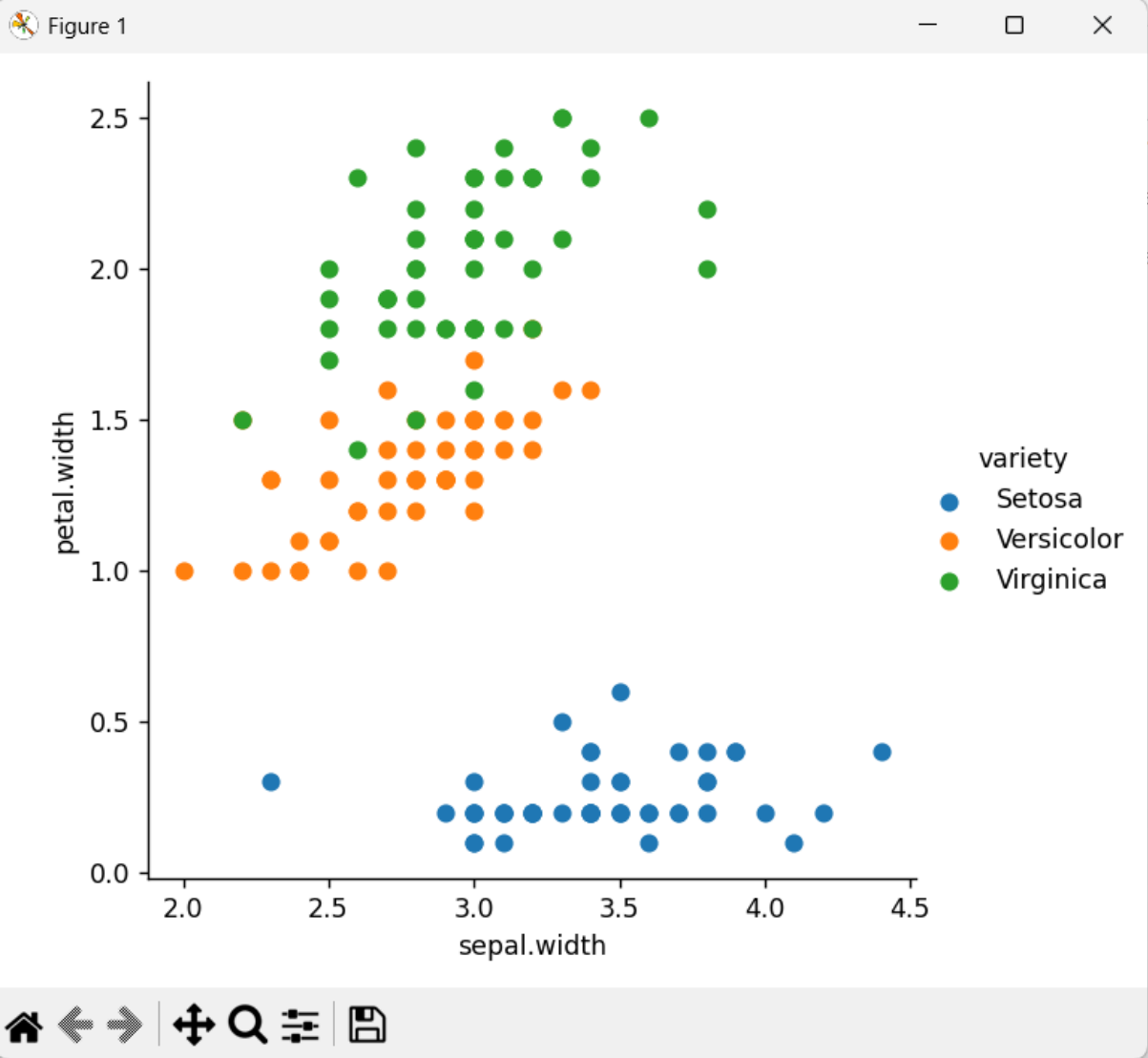
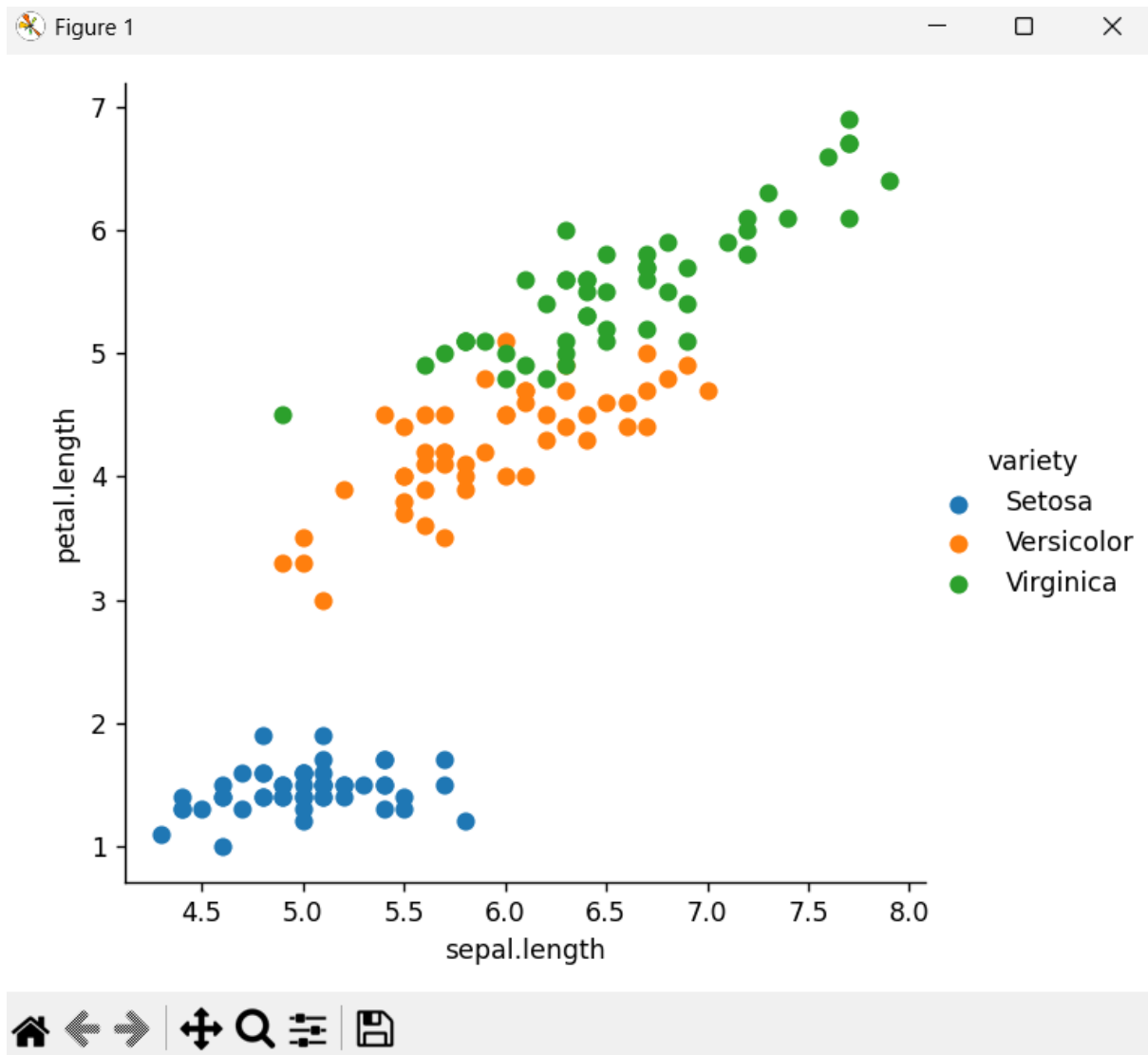


Figure 1







RESULT:

TO IMPLEMENT A PYTHON PROGRAM TO IMPLEMENT UNIVARIATE ,BIVARIATE AND MULTIVARIATE REGRESSION AS BEEN ANALYSED AND VERIFIED